A Multi-Model, Multi-Embedding Approach for Signal Peptide Classification and Intrinsically Disordered Region Asphericity Regression

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Abstract

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1 Introduction

Recent advances in deep learning have revolutionized the field of protein engineering, enabling unprecedented abilities to predict protein properties and functions directly from amino acid sequences [1]. The integration of machine learning approaches, particularly deep neural networks, has transformed traditional protein design workflows by offering rapid, cost-effective computational methods for protein characterization and optimization [2]. These computational approaches have become increasingly crucial as they significantly reduce the need for expensive and time-consuming experimental validation while maintaining high accuracy in predicting various protein properties [3]. The emergence of protein language models and advanced embedding techniques has further enhanced our

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ability to extract meaningful features from protein sequences, leading to more accurate predictions of structure-function relationships [4].

Signal peptides and protein asphericity represent two critical aspects of protein engineering that have garnered significant attention in recent years. Signal peptides, acting as molecular zip codes at protein N-termini, play a crucial role in protein trafficking and secretion, with direct implications for recombinant protein production efficiency [5]. Traditional experimental methods for signal peptide identification, such as N-terminal sequencing and fusion protein approaches, are being increasingly supplemented by machine learning-based predictors like SignalP and DeepSig [6]. Similarly, the field of intrinsically disordered proteins (IDPs) has seen remarkable progress in understanding asphericity—a key parameter describing the conformational ensemble of disordered regions. While molecular dynamics simulations have been the gold standard for calculating asphericity, their computational intensity has spurred the development of more efficient machine learning approaches, such as ALBATROSS [7], which can rapidly predict these properties directly from sequence information.

2 Methods

2.1 Dataset curation

The first task, project 1A, was centered around the classification of signal peptides. To aide in exploring signal peptides a dataset was curated using ProteinBERT and SignalP 6.0. This original data set included 4,152 rows of data, with the first column being a sequence and the second column being a 1.0 (signal peptide present) or a 0.0 (signal peptide absent). The second task, project 1B, was on the regression of asphericity. A dataset was provided that consisted of 7,462 sequences of Intrinsically Discorded Regions (IDR), along with a given asphericity value.

Despite the slight differences in data, both sets were preprocessed the same way. After receiving the raw data we applied several layers of filtering to ensure that we only maintained data that would be usable towards our modeling. The first step was ensuring that only valid amino acids were included in each sequence. Next we removed "corrupted" labels from the classification column since we were told they were artificially put there and not actually invalid sequences. Finally we ensured there were no repeated sequences, and added an additional column numerically labeling each sequence which would help with clustering. All the filtering reduced the signal peptide dataset from 4,152 rows to 3,565, and the sphericity dataset from 7,462 to 6,074.

Following processing, the data was clustered so that we ensured diversity between the train test splits. We applied clustering using mmseqs easy cluster, with a minimum sequence identity threshold of .3, a cluster coverage threshold of .5, and a coverage mode 0 which uses the shortest sequence coverage. For the classification task, the data was separated by it's class before performing clustering on each set and recombining, while no separation was necessary on the regression task. For the classification task we applied a 65-15-20 split, and used a 60-20-20 split for the regression task. For the classification task we used 65 for the training set instead of 60 since there was less total data in the classification task.

2.1.1 General Embeddings

We implemented 6 total embeddings in the project, using one hot, and ESM embeddings in both 1A and 1B. Additionally we used LM embeddings in 1A, and finally BLOSUM62 in 1B. The one hot embeddings were created from the list of 20 valid amino acids, assigning a 1 to the location in the vector corresponding to the amino acids position in our list. The sequences were normalized through padding/truncating such that they had a length of 70, making each flattened one hot encoding have a length of 1400. We used ESM embeddings as the feature rich embedding for both projects. We went with a smaller esm2 model the pretrained t68M variant which had 6 transformer layers and 8 million parameters. Using this pretrained model with each sequence we extracted a 320 length embedding of each sequence from the 6th layer of the ESM2 model.

2.1.2 1A LM embeddings

For our basic logistic regression and MLP experiments, we used ESM-2 embeddings. For our advanced model, we used 1024-dim ProteinBERT embeddings. ProteinBERT is trained on reconstruction loss, so carries a representation of the properties of each individual amino acid – we reason

that this may be a useful property in prediction signal peptides due to their strong locality to the N-terminus of the protein. As such, we made use of both the per-amino-acid embeddings and the class token embedding, an overview embedding more akin to ESM-2.

2.1.3 1B BLOSUM embeddings

The BLOSUM (BLOcks SUbstitution Matrix) matrix is a widely used substitution matrix in bioinformatics. It is a normalized scoring matrix that quantifies the likelihood of one amino acid being substituted by another during evolution. BLOSUM62, in particular, is the matrix built using sequences with less than 62 percent similarity [8]. While the BLOSUM62 matrix does not represent the inherent properties of each amino acid, using BLOSUM62 for the embedding of protein sequences for deep learning enables interpretable representation of the biochemical similarities between amino acids that enable their evolutionary substitutions. To implement the BLOSUM62 embedding, the 20*20 matrix was manually constructed and converted to a PyTorch embedding matrix to embed each amino acid into its corresponding BLOSUM62 vectors.

2.2 Architecture

2.2.1 1A architecture

The first model we implemented for the classification task was a 1-layer linear regression model. As such, the models goal was to learn a single weight vector (of equal length to the input embedded sequence) along with a bias term. The model had a single output and employed the Binary cross entropy loss function with logits loss which internally applies the sigmoid function squashing the final output to a value between 0 and 1. We employed the AdamW optimizer with varying weight decays and learning rates which will be discussed in the results section.

As for the advanced model, we used a combination of MLPs fed the ProteinBERT and Doolittle embeddings, with the AA BERT embeddings being pooled through a set of different modules per model variant. This included static elementwise weightings, learned elementwise weightings (parameterized as Gaussian), and an attention module.

2.2.2 1B architecture

* MLP
The first model we implemented for the regression task was a multilayer perception with 3 hidden layers. The first hidden layer has 128 neurons, the second hidden layer has 32 neurons, and the final hidden layer has 8 neurons. Finally we output to a single neuron to complete our regression model. We used the L1 (mean absolute error) as our loss function, and AdamW as our optimizer.

* Random Forest
The second model we implemented for the regression task was a Random Forest Regressor. We used the `RandomForestRegressor` from `sklearn.ensemble` with 100 decision trees ($\mathtt{n}_e stimators = 100$).$Eachtreewastrainedonabootstrappedsubsetofthedatatoreducevarianceandimprovegeneralization.Wesetafixe$

2.3.1 1A metrics

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The 4 evaluation metrics we used for the classification task were accuracy,
f1 score, AUROC, and MCC. Accuracy describes the overall correctness of
our model by dividing the correct predictions by the total predictions.
This serves as a good first indicator of the quality of our classification
model, however it may lack insight on the robustness of our model based on
the ratio of signal to non signal peptides in our test set.  The f1 score
helps provide more insight into the quality of our model by taking the
harmonic mean of precision and recall.  The AUROC helps provide further
insight that is not displayed in the acaccuracyy taking the area under
the curve of a plot of the true positive rate vs the false positive rate.
This test is especially important when we have an imbalanced dataset, like
the one provided for this classification task.  Finally, we have the MCC,
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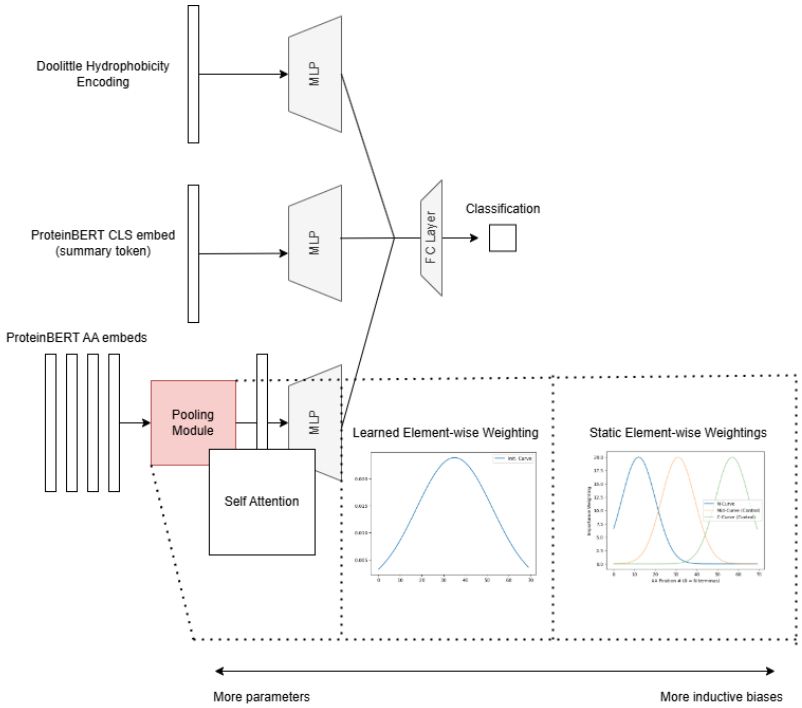


Figure 1: Advanced MLP architecture for signal peptide classification. Pooling module compared between three options.

that is also especially useful for imbalanced datasets, which measures the correlation between the true labels and predictions.

2.3.2 1B metrics

The 2 evaluation metrics we used for the regression task were Mean Squared Error (MSE) and R-squared. MSE measures the average squared difference between the predicted values and the actual values and penalizes larger errors in prediction. R-squared is also known as the coefficient of determination, which measures how well the model explains the variability of the dependent variable based on the independent variables. Calculating the R-squared values helps us determine the percentage of the variability within protein asphericity values was explained by our models.

3 Results

3.0.1 1A

Irregardless of the model used, we saw a big difference in results based on the embeddings. On a simple linear regression model, 1 layer perceptron, we achieved decent accuracy for both the one hot and VHSE embeddings. With the one hot embedded testing set having an accuracy of .861, and the VHSE embedding having an accuracy of .857. The esm embeddings were much better with an accuracy of .977. Although the accuracy was relatively high for all the embeddings, part of that can be attributed to the imbalanced data set we were working with. Looking at the MCC score, which accounts for imbalance, the one hot embeddings had a score of only .482 and the vhse embeddings had a score of .392. Both of these scores show that the model was able to correctly predict results better than randomly, since they were

above 0, however they were still not very strong. The ESM embeddings MCC score was much better at .8966.

For the merged embedding MLP model, we found that attention was the best performing model - this is interesting, because the data does seem to be in the overparameterized regime, but is likely explained by the vastly superior ability to alter the values in the embeddings (where the others can only weight an area upwards or downwards). Surprisingly, we found the C-weighted curve to slightly outperform the N-weighted one, and greatly outperform the middle-based one. However, the learned curve was comparable, and in our investigations (see below) appeared to closest to the N-based curve. This suggests that there may be an unknown benefit to the C terminus which is similar to a correctly centered N-terminus.

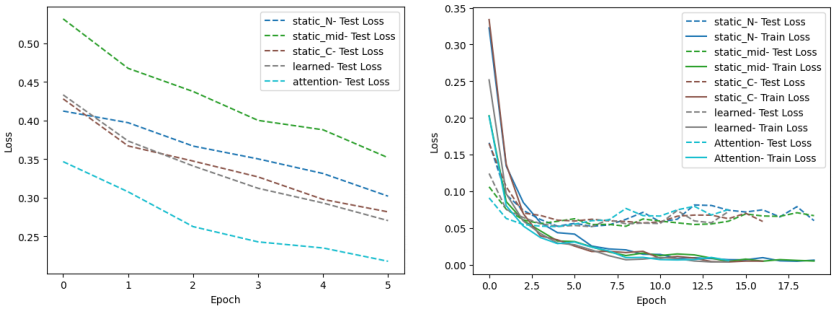


Figure 3: Comparison of losses of all different AA combining modules models. All models trained on both the CLS and AA embeddings. Note that curves converge to similar values over time (overfit regime).

We found the learned static embedding to be very vulnerable to vanishing gradients, perhaps due to having far fewer (2) parameters. While this could be counteracted by increasing the learning rate of just those parameters (while keeping the other's learning rates constant), the slowly increasing expressivity of the model revealed the model's gradual "squashing" of the Gaussian towards the N-terminus of the protein, in line with the original motivating hypothesis for a N-terminus focused curve.

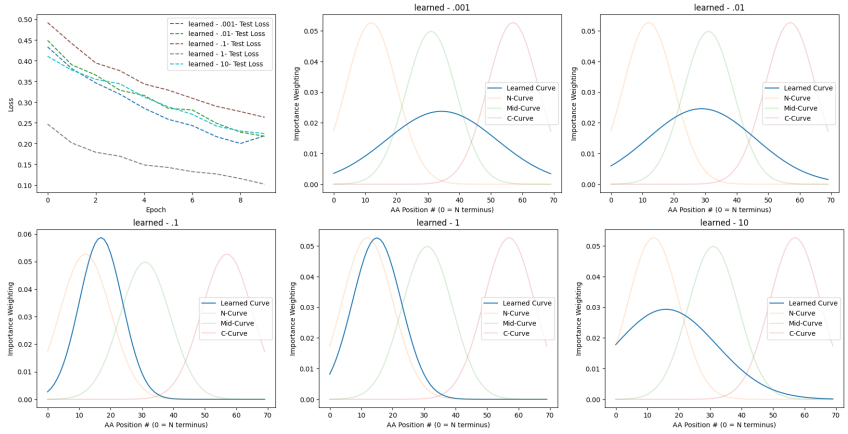


Figure 4: Test loss of the learned elementwise embedding weighting module under different learning rates. As learning rates increase, performance generally improved, and the curve was shifted to the left.

The reason we saw similar results in VHSE and onehot embeddings, but significantly better results using ESM embeddings is due to how

they represent protein sequences respectively. Both onehot and vhse lack context awareness such that each amino acid is an independent entity, with one hot being a dummy embedding and vhse providing some information about the amino acid. Though VHSE adds additional information that one hot embeddings due not they are still predefined numerical representations. Meanwhile the ESM embedding is able to capture deep contextual relationships within the sequence, as its built on a transformer model trained on millions of proteins. This allows it to learn the interactions between different amino acids and better understand structural patterns making it better than onehot and vhse.

	One-hot	VHSE	ESM
Logistic Regression	Accuracy: 0.8614 F1 Score: 0.5650 AUROC: 0.8704 MCC: 0.4827	Accuracy: 0.8571 F1 Score: 0.4624 AUROC: 0.8893 MCC: 0.3919	Accuracy: 0.9729 F1 Score: 0.9082 AUROC: 0.9906 MCC: 0.8948
MLP	Accuracy: 0.8757 F1 Score: 0.4596 AUROC: 0.9200 MCC: 0.4378	Accuracy: 0.8700 F1 Score: 0.5473 AUROC: 0.9048 MCC: 0.4751	Accuracy: 0.9729 F1 Score: 0.9108 AUROC: 0.9889 MCC: 0.8955

Figure 5: Results for Classification

3.0.2 1B

The differences in embeddings generated very little differences in prediction accuracies for the same MLP model. It is likely because the different embeddings did very little to tackle the inherent inability of MLP to predict protein asphericity levels that are very close to each other. ESM2 embedding actually generated slightly worse MSE and R-squared values compared to one-hot and BLOSUM62 embedding although it was likely not indicative of the actually fidelity of these embeddings.

Random forest regression vastly outperformed the MLP for the task of predicting protein asphericity. Its predicted values based on BLOSUM62 and ESM2 embeddings achieved an R-squared value vastly higher than those achieved by predictions made by MLP. Within predictions made by random forest regression, BLOSUM62 embeddings only marginally outperformed one-hot embeddings, with a slight increase of R-squared from 0.58020 to 0.58682, possibly due to both models having the problem of not representing any actual biochemical properties of the protein sequences. Random forest regression with ESM2 embeddings produce the best predictions out of all embeddings, possibly due to the more information provided by the ESM2 transformer.

MLP	MSE	R-squared
One-hot	0.00365	0.05064
BLOSUM62	0.00368	0.04214
ESM2	0.00375	0.02362

Figure 6: Advanced MLP architecture for Intrinsically Disordered Asphericity regression. Pooling module compared between three options.

4 Conclusion

In this study, we demonstrated the effectiveness of various embedding techniques and model architectures for both signal peptide classification and protein asphericity prediction. Our results revealed that ESM embeddings consistently outperformed traditional approaches like one-hot encoding and VHSE representations, achieving an MCC score of 0.8966 for signal peptide classification. The attention-based model showed particular promise, though its superior performance may partly result from overparameterization. One significant limitation of our work was the imbalanced nature of our signal peptide dataset, which potentially inflated accuracy metrics. Additionally, our random forest regression models, while effective, lacked interpretability regarding which sequence features most strongly influenced asphericity predictions. The relatively small dataset size for the asphericity prediction task (6,074 sequences) may have also limited our models' ability to capture the full complexity of protein conformational properties.

Future investigations should focus on several key areas we weren't able to address. First, exploring the relationship between signal peptide prediction accuracy and sequence length could help understand why some predictions fail. We observed intriguing patterns in the C-terminal weighting that merit further investigation, particularly regarding its similarity to N-terminal effects. Follow-up studies could include: (1) developing an integrated model that simultaneously predicts both signal peptides and asphericity, potentially uncovering relationships between these properties; (2) incorporating evolutionary conservation data through multiple sequence alignments to improve prediction accuracy; and (3) conducting ablation studies on the attention mechanism to better understand which sequence regions contribute most significantly to predictions. Additionally, validating our models' predictions experimentally on a set of designed proteins would provide crucial real-world performance metrics

4.0.1 1A

As for the advanced model, we found that there were signs of the benefits of locality, particularly in the model's learning of the parameters of Gaussian focus on the data, but not as clear as was initially expected, with the value of the N-terminus remaining in question. Future work may visualize the 'curve' attended to by the attention-pooling module, or focus on prying apart the differences between the CLS and AA token benefits.

References

References follow the acknowledgments in the camera-ready paper. Use unnumbered first-level heading for the references. Any choice of citation style is acceptable as long as you are consistent. It is permissible to reduce the font size to small (9 point) when listing the references. Note that the Reference section does not count towards the page limit.

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